hdoc day 45 by trishil chauhan

Menu-Driven C Program: Date Formatting and Substrings

Aim: Write a menu-driven C program using separate user-defined functions to convert a date from dd/mm/yyyy to dd-mmm-yy, print every non-empty contiguous substring of a string, and exit.

Complete C program

```
#include <stdio.h>
#include <string.h>

#define MAX 200

void changeDateFormat(void) {
    int day, month, year;
    char extra;
    const char *months[] = {"Jan", "Feb", "Mar", "Apr", "May", "Jun",
                            "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"};
    char date[40];

    printf("Enter date (dd/mm/yyyy): ");
    if (fgets(date, sizeof(date), stdin) == NULL) return;

    if (sscanf(date, "%d/%d/%d %c", &day, &month, &year, &extra) != 3 ||
        year < 1 || year > 9999 || month < 1 || month > 12) {
        printf("Invalid date.\n");
        return;
    }

    int days[] = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
    if (year % 400 == 0 || (year % 4 == 0 && year % 100 != 0))
        days[1] = 29;
    if (day < 1 || day > days[month - 1]) {
        printf("Invalid date.\n");
        return;
    }
    printf("Formatted date: %02d-%s-%02d\n",
           day, months[month - 1], year % 100);
}

void printSubstrings(const char str[]) {
    size_t n = strlen(str);
    if (n == 0) {
        printf("The string is empty.\n");
        return;
    }
    printf("All substrings:\n");
    for (size_t i = 0; i < n; i++) {
        for (size_t len = 1; len <= n - i; len++) {
            printf("%.*s", (int)len, str + i);
            if (i + 1 < n || len < n - i) printf(", ");
        }
    }
    printf("\n");
}

int main(void) {
    char input[40], str[MAX];
    int choice;
    while (1) {
```

```
        printf("\n--- STRING OPERATIONS MENU ---\n");
        printf("1. Change date format\n");
        printf("2. Print all substrings\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        if (fgets(input, sizeof(input), stdin) == NULL) break;
        if (sscanf(input, "%d", &choice) != 1) {
            printf("Invalid choice.\n");
            continue;
        }
        switch (choice) {
            case 1:
                changeDateFormat();
                break;
            case 2:
                printf("Enter a string: ");
                if (fgets(str, sizeof(str), stdin) == NULL) break;
                str[strcspn(str, "\n")] = '\0';
                printSubstrings(str);
                break;
            case 3:
                printf("Exiting program.\n");
                return 0;
            default:
                printf("Invalid choice. Try again.\n");
        }
    }
    return 0;
}
```

Sample output

```
--- STRING OPERATIONS MENU ---
1. Change date format
2. Print all substrings
3. Exit
Enter your choice: 1
Enter date (dd/mm/yyyy): 12/10/2026
Formatted date: 12-Oct-26

Enter your choice: 2
Enter a string: abc
All substrings:
a, ab, abc, b, bc, c

Enter your choice: 3
Exiting program.
```

Explanation

changeDateFormat() validates the day, month and year (including leap years) and uses a month-name array to print the requested format. printSubstrings() uses nested loops to print all contiguous substrings in start-position order. main() repeatedly displays the menu and calls the appropriate function.